Internship Assignment Optimization System

Your Name

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Management Summary

Client and Context

This project was developed for the University of Passau, specifically for the *Zentrum für Lehrkräftebildung und Fachdidaktik* (Center for Teacher Education and Subject Didactics), *Praktikumsamt für Grund- und Mittelschule* (Internship Office for Elementary and Middle Schools).

Each year, this unit is responsible for planning and organizing teaching internships for university students in collaboration with elementary and middle schools throughout the region. This planning process is time-consuming and complex, requiring careful coordination between student requirements, school capacities, and available supervising teachers. The office sought an automated solution to streamline this process while maintaining the quality and feasibility of internship assignments.

Core Challenges

The internship planning process faces several significant challenges:

- **Large number of students:** Hundreds of students require placements each semester, making manual planning impractical
- **Different internship types and subjects:** Multiple types of internships with varying subject requirements must be coordinated
- **Complex constraints:** Numerous rules govern valid assignments, including teacher qualifications, workload limits, and school type matching
- **Distance and accessibility:** Geographic considerations affect both student convenience and teacher availability
- **Overall budget constraints:** The total number of active internship placements is limited by available resources

Solution Delivered

A comprehensive web-based software system that:

- **Automates assignment optimization:** Systematically assigns teachers and students to internship placements while respecting all constraints

- **Multi-phase planning process:** Separates teacher assignment from student assignment, reflecting real-world planning workflow
- **Configurable parameters:** Allows adjustment of budget limits, constraint priorities, and optimization goals
- **Interactive management interface:** Provides tools for data input, assignment review, and manual adjustments when needed

Outcomes

The system delivers tangible benefits to the internship planning process:

- **High-quality, feasible assignment plans:** Produces assignments that satisfy hard constraints while optimizing for quality preferences
- **Customizable solutions:** Maintains flexibility for manual adjustments and what-if scenario planning
- **Reduced manual effort:** Significantly decreases planning time once initial data is entered

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Chapter 1

Introduction

1.1 Teaching Internships in Teacher Education

Practical teaching experience is a cornerstone of teacher education programs. While theoretical coursework provides foundational knowledge in pedagogy, child development, and subject-specific didactics, it is through hands-on classroom experience that future teachers develop essential professional competencies. Teaching internships bridge the gap between academic study and professional practice, offering students the opportunity to observe experienced educators, experiment with teaching methods, and receive mentored feedback in authentic school settings.

At the University of Passau, education students pursuing elementary and middle school teaching credentials are required to complete four distinct internship types throughout their program of study:

- **PDP I** (*Pädagogisch-didaktisches Praktikum I*): An initial pedagogical-didactic internship introducing students to classroom observation and basic teaching interactions
- **ZSP** (*Zusätzlich studienbegleitendes Praktikum*): An additional internship accompanying coursework, focusing on subject-specific teaching methods
- **PDP II** (*Pädagogisch-didaktisches Praktikum II*): An advanced pedagogical-didactic internship with increased teaching responsibilities
- **SFP** (*Studienbegleitendes fachdidaktisches Praktikum im Unterrichtsfach*): A subject-didactic internship in a specific teaching subject, integrating theory and practice

Each internship type serves distinct pedagogical goals and occurs at different stages of the degree program. Successful completion of all four internships is mandatory for graduation and licensure as a teacher. Critically, these internships take place in real elementary and middle schools throughout the region, supervised by active classroom teachers who mentor students and evaluate their performance.

1.2 Scale and Complexity of Internship Coordination

The logistical challenge of organizing these internships is substantial. For the 2025/2026 academic year, the *Praktikumsamt* must coordinate approximately **420 planned internship placements** across roughly **100 schools** distributed among **8 regional school districts** (*Schulämter*). Each school employs multiple teachers with varying subject specializations, availability, and capacity to supervise student interns.

The assignment process must account for numerous factors:

- **Student requirements:** Matching each student's internship type, subject specialization, and school type
- **Teacher qualifications and preferences:** Supervising teachers have individual preferences regarding which internship types and subject areas they wish to supervise
- **School characteristics:** Distinguishing between elementary schools (*Grundschule*, GS) and middle schools (*Mittelschule*, MS)
- **Student geographic accessibility:** For PDP I and PDP II internships, students must be assigned to schools close to their home addresses to minimize commute time
- **Capacity constraints:** Respecting internship capacity limits and institutional budget restrictions on the total number of active placements

1.3 Limitations of Manual Planning

Historically, internship assignments have been managed through manual processes involving spreadsheets, email coordination, and institutional knowledge. While this approach has functioned adequately in the past, it presents significant limitations:

- **Time intensity:** Manually matching hundreds of students to appropriate placements requires weeks of dedicated effort by administrative staff
- **Suboptimal assignments:** Without systematic optimization, manual assignments may inadvertently result in long commutes for students, unbalanced teacher workloads, or missed opportunities for better matches
- **Error susceptibility:** Human planners may overlook constraint violations (e.g., assigning a teacher beyond their capacity, mismatching subject requirements) or introduce data entry errors
- **Lack of transparency:** It is difficult to explain why particular assignments were made or to demonstrate that assignments are fair and equitable
- **Inflexibility:** When circumstances change (e.g., a teacher becomes unavailable, enrollment projections shift), revising assignments manually is labor-intensive and may cascade into broader disruptions

- **Limited optimization:** Manual processes struggle to balance competing objectives such as minimizing student travel distance while preserving teacher assignment continuity across semesters

These limitations motivated the *Praktikumsamt* to seek an automated solution that could systematically generate high-quality assignments while reducing administrative burden.

1.4 Course Context

This project was developed as part of the **Advanced Software Product Development (ASPD)** course at the University of Passau during the winter semester of 2025/2026. The ASPD course is structured around real-world client projects, providing students with practical experience in collaborative software engineering.

The course operates with the following organizational structure:

- **Four student teams:** Each team consists of five students, for a total of 20 participants
- **Shared client need:** All four teams address the same problem domain (internship planning), but may approach implementation differently or use different technologies
- **Agile methodology:** Development follows an iterative, sprint-based approach with **five sprints**, each lasting **two weeks**, for a total development period of ten weeks
- **Teaching assistant supervision:** Each team is guided by a dedicated teaching assistant (TA) who provides mentorship, oversight, and coordination with the client
- **Client engagement:** Communication with the *Praktikumsamt* ensures that requirements are grounded in real needs and that delivered solutions are practically deployable

This structure creates a realistic software development environment where students must balance technical challenges, evolving requirements, team coordination, and stakeholder expectations—mirroring professional software engineering practice.

1.5 Team Contribution

Our team developed a complete, standalone internship assignment system. While all four teams in the course addressed the same client need, each team worked independently to create their own implementation from scratch. Our specific approach and contributions included:

- **Constraint modeling:** Translating the complex business rules governing valid assignments into formal hard and soft constraints

- **Multi-phase optimization architecture:** Designing a phased approach that separates teacher assignment from student assignment to manage computational complexity
- **OptaPlanner integration:** Implementing constraint satisfaction solving using the OptaPlanner framework
- **Score calculation logic:** Developing efficient algorithms to evaluate solution quality based on distance minimization, workload balance, and assignment diversity
- **Backend API development:** Creating RESTful endpoints for data management, solver invocation, and result retrieval
- **Frontend interface:** Building a complete web-based user interface for system interaction

Chapter 2

Requirements Overview

2.1 System Vision

The internship assignment system is designed to support the *Praktikumsamt's* annual planning cycle while maintaining flexibility and quality. The core vision rests on three pillars:

- **Systematic assignment optimization:** Rather than relying on manual processes, the system applies algorithmic techniques to systematically generate assignments that satisfy constraints and optimize for quality
- **Real-world pragmatism:** Assignments must respect practical constraints including teacher availability, student location preferences, and institutional budgets. The system prioritizes *solution quality* over exhaustive feasibility guarantees, acknowledging that in complex problems with many competing objectives, perfect feasibility may be unattainable
- **Operational transparency:** The system maintains an audit trail of assignments and provides clear justification for decisions, enabling the *Praktikumsamt* to explain assignments to stakeholders and make adjustments when needed

2.2 Stakeholders

2.2.1 Primary: Internship Coordination Unit (*Praktikumsamt*)

The internship office is the primary stakeholder and system user. They require:

- Efficient tools to coordinate hundreds of placements annually, reducing time spent on manual assignment work
- Flexibility to review and manually adjust assignments when business circumstances change

2.2.2 Secondary: Teaching Students

Students require high-quality placements that serve their educational development. Their interests include:

- Geographic accessibility: for PDP I and PDP II internships, assignment to schools near their home addresses to make regular school visits feasible
- Subject-appropriate placements aligned with their intended teaching specialization

2.2.3 Secondary: Schools and Supervising Teachers

Schools and their supervising teachers are the infrastructure that enables internships. Their interests include:

- Balanced workload: not overwhelming individual teachers with too many interns while ensuring adequate supervision
- Alignment with subject expertise and teaching preferences (teachers have preferences for which internship types and subjects to supervise)
- Respect for institutional capacity constraints and available supervision time

2.3 Core Problem Definition

The internship assignment problem can be formally defined by specifying inputs, constraints, and objectives.

2.3.1 Input Data

The system operates on four categories of input:

1. **Students:** Each student record contains:
   - Internship type requirement (PDP I, PDP II, ZSP, or SFP for a given semester)
   - Subject specialization (primary teaching subject)
   - Home address (for distance calculations)
2. **Teachers:** Each teacher record contains:
   - Subject expertise and qualifications
   - Preferences: which internship types and subjects they are willing to supervise
   - Maximum internship capacity (workload limit per semester/year)
   - Availability constraints
3. **Schools:** Each school record contains:
   - Type: elementary (*Grundschule*, GS) or middle school (*Mittelschule*, MS)
   - Geographic location (address and coordinates)
   - Zone and public transport accessibility

- List of supervising teachers and their availability

4. **Institutional constraints:** System-level parameters including:
    - Total budget: maximum number of active internship placements for the semester/year

2.3.2 Constraints

The system must respect multiple categories of constraints representing real-world business rules and stakeholder requirements:

Hard Constraints (must be satisfied)

Hard constraints are non-negotiable requirements reflecting institutional policies and operational feasibility:

- **Budget constraint:** The total number of scheduled internship placements cannot exceed the institutional budget limit for the academic year
- **Complete student assignment:** Every student must receive exactl internship placement matching their internships requirements
- **Teacher workload limits:** Each teacher can supervise 0, 1, 2, or 4 internships per planning cycle (semester or year, depending on internship type).[1]
- **Teacher max workload:** Each teacher can specify a max number of internships they prefer to not exceed
- **Internships capacities:** PDP internships can accommodate at maximum two student, SFP and ZSP can accommodate at maximum 4
- **Teacher diversity requirement:** If a teacher supervises multiple internships, they must be of different types. A teacher cannot supervise multiple instances of the same internship type (e.g., two PDP I internships)
- **Subject qualification:** Teachers supervising ZSP or SFP internships (which are subject-specific) must have relevant subject expertise. SFP internships must match the student's main course
- **School type matching:** Students pursuing elementary school certification must be assigned to elementary schools (GS), and those pursuing middle school certification to middle schools (MS)
- **School zones matching:** Internships, by type, must take place in schools with specific zones
- **Teacher availability:** Internship placements can only be scheduled at schools with available, qualified supervising teachers

[1] The *Praktikumsamt_Uebersicht* documentation specifies that teachers should be assigned exactly 2 internships. However, analysis of historical data (example_data) reveals teachers with 1, 2, and 4 internship assignments, with 2 being the most common configuration. The system accommodates this flexibility while treating 2 internships as the preferred workload through soft constraint optimization.

Soft Constraints (preferences to optimize)

Soft constraints are desirable properties that improve assignment quality and stakeholder satisfaction:

- **Student proximity for PDP I & PDP II:** Minimize the geographic distance between assigned students and their home addresses. This ensures practical feasibility for repeated school visits
- **Teacher workload preference:** Prefer assigning teachers exactly 2 internships over other valid counts (1, 4, or 0). This represents ideal teacher engagement
- **Diversity at schools:** Prefer schools with diverse internship types, rather than schools where one type dominates
- **Teacher preference matching:** Favor assignments of internship types and subjects that align with individual teacher preferences
- **Teacher assignment preservation:** When reoptimizing across semesters, prefer preserving teacher assignments for the same types of internships. This promotes continuity and relationship-building between teachers and the university program
- **Subject Matching for ZSP:** Unlike SFP, ZSP internships are flexible regarding the subject course, favor assignments that align with students preferences (main course + 3 additional preferred courses)

2.3.3 Objectives

The system has two primary objectives:

1. **Feasibility:** Generate assignment plans that satisfy all hard constraints. This ensures that assignments are valid and implementable
2. **Optimization:** Among feasible solutions, select plans that maximize soft constraint satisfaction. This means minimizing student travel distances, balancing teacher workloads, respecting preferences, and achieving other quality goals

Importantly, **this system is an optimization system rather than a pure feasibility solver**. In problems with many competing constraints and objectives, guaranteed feasibility across all scenarios may be unattainable. The system prioritizes generating *high-quality, near-feasible* solutions over mathematical certainty of feasibility. This pragmatic approach reflects real-world practice: even if 100% feasibility cannot be assured, the system can produce solutions that satisfy most constraints and provide a high-quality starting point for manual adjustment if needed.

Feasibility Landscape

A key observation mitigates feasibility concerns: the institution has a favorable constraint-to-resource ratio. The total number of available supervising teachers substantially exceeds the number of students requiring placements. For the 2025/2026 academic year, approximately 100 schools with multiple teachers each, must accommodate roughly

420 planned internships. This surplus of teacher availability significantly improves the likelihood of finding feasible solutions in most scenarios.

However, the system is deliberately structured to strategically satisfy constraints even when full satisfaction is impossible, rather than assuming feasibility is guaranteed. This defensive design approach:

- Acknowledges that specific combinations of constraints (e.g., specialized subject requirements + teacher preferences) can create local infeasibility despite overall abundance
- Implements constraint prioritization so that violations, if they must occur, affect lower-priority (preference) constraints rather than critical hard (impossibility) constraints
- Ensures robust behavior even in edge cases or future scenarios with tighter resource availability

2.4 System Characteristics

2.4.1 Deterministic Optimization

The system uses constraint satisfaction and local search optimization to explore the solution space. While it does not guarantee globally optimal solutions (such problems are NP-hard), it consistently produces high-quality solutions within acceptable runtime limits.

2.4.2 Configurability

The *Praktikumsamt* can adjust:

- Budget limits per semester
- Teacher preferences and availability

2.4.3 Interactive Assignment Management

While the core optimization engine operates automatically, the system provides interfaces for:

- Reviewing and validating proposed assignments
- Making manual adjustments when needed (e.g., if a teacher becomes unavailable)

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